

* Datatypes

→ A datatype is classification that tell a computer what kind of value a variable holds and what operations can be performed on what data.

or

A datatype defines the type of data a variable can store such as numbers text or logical values.

Two types

- Single valued DT
 - Int • float • Complex • Boolean
- Multi valued DT
 - str • list • tuple • set • dictionary is also called as data structures.

Ex:

a = 25

b = 2.58

type(a) # Int type(b) # float

- In Initializing the value to variable we have some methods

MI • a = 2 c = 9

b = 10 print(a, b, c) # 2 10 9

M2 $x = 4$; $y = 15$; $z = 45$ `print(x, y, z)` # 4 15 45M3 $m, n, p = 12, 45, 67$ `print(m, n, p)` # 12 45 67prgm

Write a program to swap a number
 O/P | before swap $a = 10$ & $b = 20$ | after
 swap $a = 20$ & $b = 10$

try with 3rd variable & without 3rd variable

with 3rd variable

 $a = 10$ $b = 20$ `print("Before swap a = {a} and b = {b}")` $temp = a$ $a = b$ $b = temp$ `print("After swap a = {a} and b = {b}")`

O/P

Before swap $a = 10$ and $b = 20$ After swap $a = 20$ and $b = 10$

without 3rd variable

 $a = 10$ $b = 20$ `print("Before swap a = {a} and b = {b}")`

$a, b = b, a$

Print ("After swap $a=2a$ and $b=2b$ ")

O/P

Before swap $a=10$ and $b=20$

After swap $a=20$ and $b=10$

prog

Write a program to calculate area of rectangle, triangle, circle

$area = l * b$

$area = 0.5 * base * height$

$area = 3.142 * r * r$

$l = \text{float}(\text{input}(\text{"Enter a len of rec: "}))$

$b = \text{float}(\text{input}(\text{"Enter a breadth of rec: "}))$

$base = \text{float}(\text{input}(\text{"Enter a base of tri: "}))$

$h = \text{float}(\text{input}(\text{"Enter a height of tri: "}))$

$r = \text{float}(\text{input}(\text{"Enter a radius of circle: "}))$

$area \text{ of rec} = l * b$

Print ("Area of Rectangle", $area$)

$tri = 0.5 * base * h$

Print ("Area of triangle", tri)

$ci = 3.142 * r * r$

Print ("Area of circle", c1)

01P Enter a length of rec : 23

Enter a breadth of rec : 12

Enter a base of tri : 45

Enter a height of tri : 24

Enter a radius of circle : 5

Area of Rectangle : 276.0

Area of Triangle : 540.0

Area of circle : 78.55